

Screening Test - Kaprekar Contest

Saturday, August 31, 2002.

- 1) Fill in the answer sheet with your Name, Class, Section etc. in the specified places.
- 2) Only one of the choices A,B,C,D is correct for each question. Write the alphabet of your choice in the answer sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).
- 3) For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{4}$ mark.
- 4) Diagrams are only visual aids; they are not drawn to scale.
- 5) You are free to do rough work on separate sheets.
- 6) Duration of the test: 2 p.m. to 4 p.m.

- $$\begin{array}{r} A \\ + B \\ + C \\ \hline A \quad B \quad C \end{array}$$

- A) 1,9,8 B) 1,8,9
C) 1,7,9 D) 1,9,7

- A) 121 B) 11 C) 2 D) 3

3. The value of $\frac{1}{3}$ of 15^{27} is

- A) 5^{27} B) 15^9 C) 5×15^{26} D) 5×3^9

4. The sum of nine consecutive whole numbers is 99. the largest of these integers is

- A) 19 B) 15 C) 17 D) 21

5. A student ranked 9th from the top and 38th from the bottom in a class. The number of students in the class is

- A) 45 B) 46 C) 47 D) 48

6. A and B are respectively 20% and 10% lower than C . What percent of B is A ?

- A) 66.66 B) 88.88 C) 50 D) 45

7. If the 15th January 1997 was a Wednesday, than 15th February 2006 will be a

- A) Wednesday B) Tuesday C) Sunday D) Friday

8. In an examination 60% passed in English, 52% passed in Maths while 32% failed in both. If 220 candidates passed in both, then the total number of candidates was

- A) 200 B) 300 C) 400 D) 500

9. If 10% of m is the same as 20% of n , then $m:n$ is

- A) 5:1 B) 2:1 C) 1:2 D) 10:1

10. For an integer n , a student states the following:

I. If n is odd, $(n+1)^2$ is even

II. If n is even, $(n-1)^2$ is odd

III. If n is even, $\sqrt{(n-1)}$ is irrational.

Which of the above would be true?

A) I & III

B) I & II

C) All

D) II & III

11. If $a=2$, $b=3a+4$ and $c=3b$, then $b-c$ is

A) 30

B) 20

C) -20

D) -22

12. The numbers 52, 32, 12, 43, 25, 18, 56, 36, 16, 50 are grouped in pairs in such a way that the sum of each pair is the same. The number paired with 18 is

A) 52

B) 50

C) 43

D) 36

13. The time on an electronic digital watch is 11.11. How many minutes before this would the watch have shown a time with all digits identical?

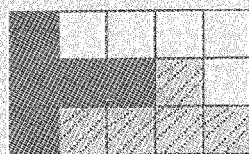
A) 72

B) 144

C) 216

D) 316

14. Observe the way the squared paper in the figure is divided into three blocks. Which of them is a net for an open cubical box?



A) X only

B) Y only

C) Y and Z only

D) All

X block is shaded

Y block is shaded

Z block is shaded

15. Let $a*b = a + b + ab$ where a, b are rational numbers, neither of them equal to -1 . For example

$$6*2 = 6+2+(6 \times 2) = 20$$

$$-2*\frac{1}{4} = -2+\frac{1}{4}+(-2 \times \frac{1}{4}) = -2\frac{1}{4}$$

If $a*x = 0$, then x is

A) $\frac{a}{a+1}$

B) $\frac{a}{a-1}$

C) $\frac{-a}{a-1}$

D) $\frac{-a}{a+1}$

16. Three boys P, Q, R agree to divide a bag of marbles as follows:

P takes one more than half of the marbles; Q takes a third of the remaining marbles;

R takes the marbles left out now in the bag.

The original number of marbles found at the beginning in the bag must be

A) a multiple of 6

B) one more than a multiple of 6

C) two more than a multiple of 6

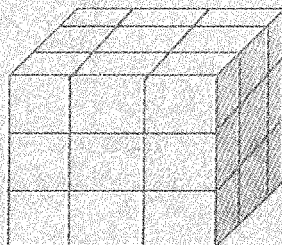
D) three more than a multiple of 6

17. The adjoining figure is formed by two squares. The side of each square is a whole number. If the area of the figure is 58 cm^2 , its perimeter is



- A) 58 cm B) 34 cm
C) 29 cm D) 116 cm

18. The large cube in the figure consists of 27 unit cubes. If all the six faces are painted in colour (say, in blue), how many unit cubes have only one (blue) coloured face?



- A) 6 B) 8
C) 12 D) 27

19. A chocolate drink is 8% pure chocolate, by volume. If 5 litres of pure milk are added to 25 litres of this drink, the percent of chocolate in the new drink is nearly

- A) 10 B) 7 C) 15 D) 13

20. If $a^2bc^3 = 5^3$ and $ab^2 = 5^6$, then abc equals

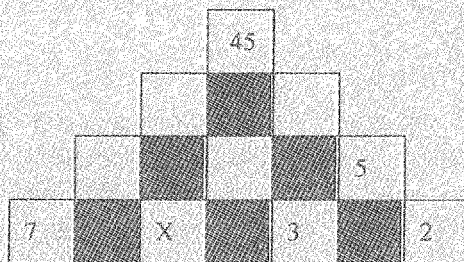
- A) 5 B) 5^2 C) 5^3 D) $5^{4.5}$

21. Here is a magic square, created using the four numbers 10, 20, 30, 40 once in each row, column or diagonal. The value of $A+B$ is

C	10	D	E
A	30	F	G
B	H	20	K
L	M	N	10

- A) 10 B) 20
C) 30 D) 40

22. The adjacent number pyramid is formed by filling up each unshaded square with the sum of the two numbers connected to it from the row below. (Thus 5 is obtained by adding 3 and 2). The top number obtained at the end is 45. The value of X is



- A) 6 B) 9
C) 17 D) 12

23. S_1 , S_2 , S_3 are three sums of money such that S_2 is the simple interest on S_1 and S_3 is the simple interest on S_2 for the same rate and same period. Then

- A) $S_1^2 = S_2 S_3$ B) $S_3^2 = S_1 S_2$ C) $S_2^2 = S_1 S_3$ D) $S_1 S_2 S_3 = 1$

24. If $x > 2y$ and $z < \frac{1}{3}$ then

A) $x > 6z$

B) $x < 6z$

C) $y < \frac{6}{x}$

D) $z > \frac{6}{x}$

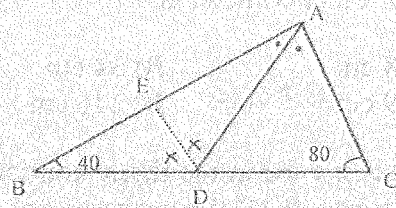
25. In the adjacent figure $\overrightarrow{DE}$ bisects $\angle BDA$ and $\overrightarrow{AD}$ bisects $\angle BAC$. The measure of $\angle BED$ is

A) 85°

B) 95°

C) 75°

D) 105°



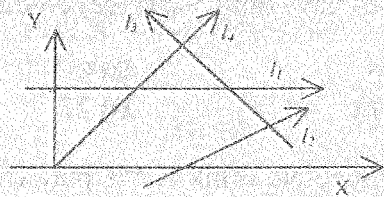
26. In the diagram, there are four lines. The line whose graph could be $y=10-x$ is

A) l_1

B) l_2

C) l_3

D) l_4



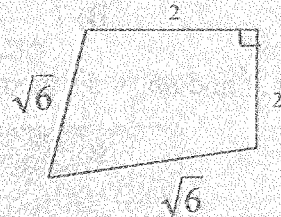
27. The area of the quadrilateral shown in the figure is

A) 6

B) $2+\sqrt{6}$

C) $2+\sqrt{8}$

D) $\sqrt{12}$



28. The altitude drawn to the base of an isosceles triangle is of length 8 and the perimeter is 32. The area of the triangle is

A) 32

B) 40

C) 48

D) 56

29. In a right angled triangle, if the square of the hypotenuse is twice the product of the other two sides, then one of the angles of the triangle is

A) 30°

B) 45°

C) 60°

D) 15°

30. A cylindrical pail containing water drains into a cylindrical tub of diameter 80 cm and height 100 cm, while resting at an angle of 45° to the horizontal, as shown. How deep is the water in the tub when its level reaches the pail?

A) 40 cm

B) 50 cm

C) 60 cm

D) 45 cm

